Analysis of Algorithms

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https://github.com/rolandogonzlz/Analysis_Algorithms/tree/main/Codes

1. Solve the following recurrence. The solution of each exercise includes finding the closed from and then verifying using mathematical induction. The solution of the recurrences can be done by hand and then scanned t

1.1 $T(n) = 3T(n/2) + n$, $T(1) = 0$ for all n powers of two $1, 2, 4, 8, 10, \ldots$

$n = 2^m$

$S(m) = T(2^m)$

$S(0) = T(1) = 0$

$S(m) = 3S(m-1) + 2^m$

$S_h(m) = C\,3^m$

$S_p(m) = a \cdot 2^m \Rightarrow -2 \cdot 2^m$

$a2^m = 3(a2^{m-1}) + 2^m$

$a2^m = \frac{3a}{2}2^m + 2^m(2^m)$

$a = \frac{3a}{2} + 1 \Rightarrow a - \frac{3a}{2} = 1$

$-\frac{a}{2} = 1 \Rightarrow a = -2$

$S(m) = C \cdot 3^m - 2 \cdot 2^m$

$S(0) = 0$

$0 = S(0) = C \cdot 3^0 - 2 \cdot 2^0 \Rightarrow C = 2$

$S(m) = 2 \cdot 3^m - 2 \cdot 2^m = 2(3^m - 2^m)$

$\boxed{n = 2^m}$

$T(2^m) = 2(3^m - 2^m)$

$m = 0$

$T(2^0) = T(1) = 0$

$2(3^0 - 2^0) = 2(1-1) = 0$

Induction step

Assume true for $m = k$

Show true for $m = k+1$

Assume: $T(2^k) = 2(3^k - 2^k)$

Show: $T(2^{k+1}) = 3T(2^k) + 2^{k+1}$

$T(2^{k+1}) = 3(2(3^k - 2^k)) + 2^{k+1}$

$= 6 \cdot 3^k - 6 \cdot 2^k + 2^{k+1}$

$= 2 \cdot 3^{k+1} - [6 \cdot 2^k - 2^{k+1}]$

$2^{k+1} = 2 \cdot 2^k$

$6 \cdot 2^k - 2^{k+1} = 6 \cdot 2^k - 2 \cdot 2^k = 4 \cdot 2^k = 2 \cdot 2^{k+1}$

$T(2^{k+1}) = 2 \cdot 3^{k+1} - 2 \cdot 2^{k+1} \Rightarrow$

$2(3^{k+1} - 2^{k+1})$

$T(n) = 2(3^m - 2^m)$ where $n = 2^m$

$T(n) = 3\,T(n-1) + 4T(n-2)$

$T(0) = 0$

$T(1) = 5$

$T(2) = 3T(1) + 4T(0) = 15$

$T(3) = 3T(2) + 4T(1) = 65$

$\vdots$

with $n=0$ $\quad A+B=0$

with $n=1$ $\quad 4A - B = 5$

$A = 1, \; B = -1$

$T(n) = r^n$

$r^n = 3r^{n-1} + 4r^{n-2} \quad (r^{n-2})$

$r^2 = 3r + 4$

$r^2 - 3r - 4 = 0$

$(r-4)(r+1)$

$r_1 = 4 \quad r_2 = -1$

$T(n) = A4^n + B(-1)^n$

$T(n) = 4^n - (-1)^n$

Induction step

Assume true for $n=k$

show true for $n = k+1$

Assume: $T(k) = 4^k - (-1)^k$

Show: $T(k+1) = 4^{k+1} - (-1)^{k+1}$

$3T(k+1-1) + 4T(k+1-2)$

$3T(k) + 4T(k-1)$

$3[4^k - (-1)^k] + 4[4^{k-1} - (-1)^{k-1}]$

$4^1 \cdot 4^{k-1} = 4^k$

$(-1)^{k-1} = (-1)^k \left(\frac{1}{(-1)}\right)$

$\Rightarrow \underline{3 \cdot 4^k} - 3(-1)^k + \underline{4 \cdot 4^{k-1}} - 4(-1)^{k-1}$

$3 \cdot 4^k + 4 \cdot 4^{k-1} \Rightarrow 3 \cdot 4^k + 4^k \Rightarrow 4^k(3+1) = 4^k \cdot 4 = 4^{k+1}$

$-3(-1)^k - 4(-1)^{k-1} \Rightarrow -3(-1)^k - 4\left[(-1)^k\left(\frac{1}{-1}\right)\right]$

$-3(-1)^k + 4(-1)^k$

$(-1)^k(-3+4) \Rightarrow (-1)^k(1) \Rightarrow -(-1)^{k+1}$

$4^{k+1} - (-1)^{k+1}$

$T(n) = 5T(n-1) - 6T(n-2)$

$T(0) = 0$

$T(1) = 1$

$T(n) = r^n$

$r^n = 5r^{n-1} - 6r^{n-2} \quad (r^{n-2})$

$r^2 = 5r - 6$

$r^2 - 5r + 6 = 0$

$(r-3)(r-2) = 0$

$r_1 = 3 \quad r_2 = 2$

$T(n) = A3^n + B2^n$

$n=0 : A + B = 0 ; B = -A$

$n=1 : 3A + 2B = 1$

$3A + 2(-A) = 1 \Rightarrow A = 1$

$B = -1$

$T(n) = 3^n - 2^n$

Induction step

Assume true for $n = k$

show true for $n = k+1$

Assume

$T(k) = 3^k - 2^k$

show:

$T(k+1) = 3^{k+1} - 2^{k+1} \Rightarrow T(k-1) = 3^{k-1} - 2^{k-1}$

$5T(k+1-1) - 6T(k+1-2)$

$5T(k) - 6T(k-1)$

$5(3^k - 2^k) - 6(3^{k-1} - 2^{k-1})$

$5 \cdot 3^k - 5 \cdot 2^k - 6 \cdot 3^{k-1} + 6 \cdot 2^{k-1}$

$\Rightarrow 5 \cdot 3^k - 6 \cdot 3^{k-1} - 5 \cdot 2^k + 6 \cdot 2^{k-1}$

$\Rightarrow 5 \cdot 3^k - 6 \cdot 3^{k-1} = 3^{k-1}(5 \cdot 3 - 6) \Rightarrow 3^{k-1}(15 - 6)$

$\Rightarrow 3^{k-1} \cdot 9 \Rightarrow 3^{k-1} \cdot 3^2 \Rightarrow \underline{\underline{3^{k+1}}}$

$\Rightarrow -5 \cdot 2^k + 6 \cdot 2^{k-1} = 2^{k-1}(-5 \cdot 2 + 6)$

$\Rightarrow 2^{k-1}(-10 + 6) \Rightarrow 2^{k-1}(-4) \Rightarrow 2^{k-1} \cdot (-2^2) \Rightarrow -2^{k+1}$

$T(k+1) = 3^{k+1} - 2^{k+1}$

#

$x(n) = 2x(n-1) + 4x(n-2)$

$x(0) = 1$ for $n > 1$

$x(1) = 2$

$x(2) = 2x(1) + 4x(0) = 2(2) + 4(1) = 4 + 4 = 8$

$x(3) = 2x(2) + 4x(1) = 2(8) + 4(2) = 16 + 8 = 24$

We now that : $x(n) = r^n$

$(r^{n-2})\ r^n = 2r^{n-1} + 4r^{n-2}$

$r^2 = 2r + 4$

$r^2 - 2r - 4 = 0$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-4)}}{2(1)} = 1 \pm \sqrt{5}$$

So: $r_1 = 1 + \sqrt{5}$; $r_2 = 1 - \sqrt{5}$

$x(n) = A(1+\sqrt{5})^n + B(1-\sqrt{5})^n$

I.C. with $n=0$; with $n=1$

$x(0) = A + B = 1$ $\quad$ $x(1) = A(1+\sqrt{5}) + B(1-\sqrt{5}) = 2$

$A = \frac{5+\sqrt{5}}{10}$ $\quad$ $B = \frac{5-\sqrt{5}}{10}$

$$x(n) = \frac{5+\sqrt{5}}{10}(1+\sqrt{5})^n + \frac{5-\sqrt{5}}{10}(1-\sqrt{5})^n$$

Induction step:

Assume true for $n=k$ show true for $n=k+1$

Assume: $x(k) = Ar_1^k + Br_2^k$

Show: $x(k+1) = Ar_1^{k+1} + Br_2^{k+1}$

$2x(k+1-1) + 4x(k+1-2)$

$2x(k) + 4x(k-1)$

$\Rightarrow 2(Ar_1^k + Br_2^k) + 4(Ar_1^{k-1} + Br_2^{k-1})$

$\Rightarrow A(2r_1^k + 4r_1^{k-1}) + B(2r_2^k + 4r_2^{k-1})$

$\Rightarrow Ar_1^{k-1}(2r_1 + 4) + Br_2^{k-1}(2r_2 + 4)$

$r^2 = 2r + 4$ so:

$2r_1 + 4 = r_1^2$; $2r_2 + 4 = r_2^2$

$\Rightarrow Ar_1^{k-1}r_1^2 + Br_2^{k-1}r_2^2$

$\Rightarrow Ar_1^{k+1} + Br_2^{k+2}$

$\Rightarrow x(k+1) = A(1+\sqrt{5})^{k+1} + B(1-\sqrt{5})^{k+1}$ //